

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used standard Python indentation (4 spaces) throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project follows modular architecture with separate folders for routers, services, models, frontend, and tests.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as <code>user_query</code> , <code>quiz_data</code> , <code>summary_text</code> , and <code>learning_path</code> are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>generate_quiz()</code> , <code>summarize_text()</code> , <code>explain_concept()</code> , and <code>recommend_learning_path()</code> are meaningful.
5	Code Comments	Inline and block comments explain logic	Yes	Major sections and functionality are documented using comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Backend logic is divided into reusable service modules and API routers.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions and modular architecture reduce code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks and FastAPI exception handling are implemented for API requests and database operations..

Reusable Components / Modules:

S.No	Component / Module Name	Language Technology	Where Reused	Reusability Level
1	Authentication Service	Python, JWT, bcrypt	User registration, login, session validation	High
2	AI Orchestrator Service	Python, Google Gemini 1.5 Flash, LaMini-Flan-T5	Question Answering, Explanation, Quiz, Summary, Learning Path	High
3	Database Service	Python, SQLAlchemy, SQLite	User management, history, bookmarks, AI responses	High
4	API Router Modules	FastAPI	Authentication, AI features, history, frontend routes	High
5	Frontend Components	HTML, CSS, JavaScript, Jinja2	Dashboard, Login, Quiz, Learning Path, Summary pages	High

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Authentication Service	Python, JWT, bcrypt
2	AI Orchestrator Service	Google Gemini 1.5 Flash, LaMini-Flan-T5
3	Database Service	SQLAlchemy, SQLite
4	API Router Modules	FastAPI
5	Frontend Components	HTML, CSS, JavaScript, Jinja2